

ArchiFlow AI

DevOps Runbook

Incident: SYNC Button Non-Responsive

1. Purpose

This runbook defines the operational steps to diagnose, contain, repair, validate, and harden the SYNC workflow in ArchiFlow AI when the SYNC button becomes non-responsive.

This document is written for: - DevOps Engineers - Backend Developers - Frontend Developers - Platform Reliability Engineers

This runbook must be followed strictly. No silent failures are acceptable.

2. System Flow Overview (Expected Behavior)

SYNC → Frontend Event → API Endpoint → ai_jobs Creation → LLM Execution → Diagram Version Creation → Rendering → UI Update → Audit Log

If any stage fails silently, the system is considered unstable.

3. Severity Classification

Severity: HIGH Reason: Core feature failure affecting diagram generation and revision.

4. Phase 1 – Immediate Triage Checklist

4.1 Confirm Environment

- Confirm environment: Production / Staging / Local
- Confirm user role (Admin / Editor / Viewer)
- Confirm LLM connection status

4.2 Browser-Level Check

Open DevTools: - Check Console errors - Check Network tab for API request - Confirm button has click event bound - Confirm request payload is sent

If no request is triggered → Frontend issue. If request is triggered but no response → Backend issue.

5. Phase 2 – Backend Investigation

5.1 API Verification

Verify endpoint exists: - POST /api/v1/projects/{projectId}/diagrams:generate - POST /api/v1/projects/{projectId}/diagrams/{diagramId}:revise

Check: - Request validation - Authentication - RBAC enforcement

5.2 Job Creation Validation

Query database:

```
SELECT * FROM ai_jobs ORDER BY created_at DESC LIMIT 5;
```

Confirm: - Job created - Status transitions (queued → running → success/failed) - error_message populated if failed

If no job created → API handler failing before orchestration.

5.3 Job Step Trace (If Enabled)

```
SELECT * FROM ai_job_steps WHERE job_id = 'JOB_ID';
```

Confirm: - LLM call step exists - Parse step exists - Render step exists

6. Phase 3 – LLM Connector Validation

6.1 Connection Status

```
SELECT * FROM llm_connections WHERE status != 'active';
```

If expired/revoked → Trigger reauthorization flow.

6.2 Provider Reachability

Check: - Provider API reachable - Timeout settings - Token expiry - Provider response format

If provider unreachable: - Return structured LLM_UNAVAILABLE error - Attempt fallback provider

7. Phase 4 – Rendering Validation

Check: - diagram_versions record created - spec_text not null - render_status not stuck in queued - render_output_asset_id created

If Mermaid parsing fails: - Capture error - Return RENDER_FAILED - Log structured failure

8. Containment Procedure

If SYNC failing in production:

1. Disable SYNC button temporarily
 2. Display maintenance message
 3. Enable detailed logging
 4. Activate job step tracing
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9. Permanent Repair Requirements

9.1 Frontend Hardening

- Mandatory loading spinner
- Disable SYNC while running
- Visible error notification
- No silent try/catch blocks

9.2 Backend Hardening

- Transactional ai_jobs creation
- Explicit error responses
- Correlation ID logging
- Status transition enforcement

9.3 LLM Hardening

- Token validation before execution
- Fallback routing
- Execution timeout handling
- Cost logging (tokens_in / tokens_out)

9.4 Rendering Hardening

- Pre-render validation
 - Structured failure codes
 - Asset pointer verification
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10. Observability Requirements

Enable: - ai_job_steps for traceability - Correlation IDs across services - Performance metrics (generation duration) - Audit logging for GENERATE_DIAGRAM

No orchestration action may occur without an audit entry.

11. Validation Checklist (Post-Fix)

1. Click SYNC
2. Spinner appears
3. API call visible
4. ai_job created
5. LLM executed
6. diagram_version created
7. render successful
8. UI updated
9. audit_event logged

Performance target: <5 seconds for standard cases.

12. Regression Testing Requirements

Automate tests for: - SYNC event binding - API success path - API failure path - LLM unavailable scenario - Rendering failure scenario - Token expired scenario

No release allowed without passing SYNC regression suite.

13. MASTER PROMPT FOR ANTIGRAVITY SYSTEM

Use the following directive when invoking system self-diagnosis:

You are the ArchiFlow AI System Recovery Agent.

The SYNC button is non-responsive.

Execute full system trace across: 1. Frontend event binding 2. API route execution 3. ai_jobs creation and status 4. LLM connector execution 5. diagram_versions creation 6. Rendering output 7. audit_events logging

For each layer, return: - Status (OK / FAILED) - Evidence (logs, records, timestamps) - Root cause if failure detected

If failure is detected: - Apply automated repair - Enforce structured error handling - Add visible user feedback - Enable job step tracing - Validate token status - Retry with fallback provider if necessary

After repair: - Re-test SYNC end-to-end - Confirm version creation - Confirm rendering - Confirm audit log - Provide performance metrics

Final output must include: 1. Root Cause 2. Layer Affected 3. Fix Applied 4. Preventive Enhancements 5. Confirmation Test Result

No silent failures are allowed. No LLM tokens may be stored server-side. All actions must be logged with correlation_id.

End of Runbook.